

1 THE FROZEN-LATTICE THEOREM

1.0.1 Intersection dimensions in the matching-subspace arrangement are frozen across the tower

1.1 Setup

Over F , index six coordinates by $\{0, \dots, 5\}$. K has 15 perfect matchings $J_1, \dots, J_{15}$. For a matching J , let $L_J = \{y_a : y_a + y_b = 0, (a, b) \in J\}$, a 3-dimensional diagonal subspace. For a nonempty set T of matchings, let $E(T) = \{(a, b) \in J : J \in T\}$ and let $c(T) = \text{number of bipartite connected components of the graph } (6 \text{ vertices, edges } E(T)) = 6 - \text{rank}\{F\}_{\{e : e \in E(T)\}}$.

At tower level v , the arrangement is $\{(L_J)^v : J \in T\}$, and the intersection over T is $(\bigcap_{J \in T} L_J)^v$.

1.2 Theorem (Frozen Lattice)

For every nonempty $T \subseteq \{1, \dots, 15\}$ and every $v \geq 1$: $\dim(\bigcap_{J \in T} (L_J)^v) = v \cdot c(T)$, hence $|\bigcap_{J \in T} (L_J)^v(F)| = 3^{v \cdot c(T)}$.

The intersection dimension per level is $c(T)$, **independent of v** (“frozen”); the tower merely multiplies it by v . In particular the numbers $3^{c(T)}$ that feed the inclusion–exclusion of the arrangement are **char-independent and v -independent** — they are set counts of a fixed finite arrangement, carrying no torsion.

1.3 Proof

The intersection $\bigcap_{J \in T} L_J$ is the solution space in F^6 of the linear system $\{y_e = 0 : e \in E(T)\}$. Its dimension is $6 - \text{rank}\{F\}_{\{e : e \in E(T)\}}$. Each $y_e = s_a + s_b$ is the (signed) incidence vector of edge e ; the rank of a set of such vectors over any field equals $|V| - (\text{number of bipartite components of the edge graph})$ — the standard rank of a graph’s incidence matrix, where a component contributes (its vertex count $- 1$) to the rank if it contains an odd cycle, and (vertex count $- 1$) with one extra relation, i.e. bipartite components each drop the rank by one relative to a spanning forest. Concretely $\text{rank} = 6 - c(T)$ with $c(T)$ the bipartite-component count, giving $\dim(\bigcap_{J \in T} L_J) = c(T)$.

For the product, $(\bigcap_{J \in T} L_J)^v = \bigcap_{J \in T} (L_J)^v$ (products commute with these coordinate-wise intersections), and $\dim$ of a v -fold product is v times the factor dimension, so $\dim = v \cdot c(T)$ and the point count is $3^{v \cdot c(T)}$. Nothing in the argument uses the field beyond $\text{char} \neq 2$ issues absorbed into the bipartite-component count; the value $c(T)$ is a graph invariant of the fixed arrangement, identical at every tower level.

1.4 Verification (byte-exact, direct enumeration at $v=1$)

T (sample)	$ L_J(F) $	$3^{c(T)}$	$c(T)$
singleton $\{J\}$ (all 15)	27	27	3
pair, bipartite-merging	3	3	1
pair, disjoint-merging	9	9	2
triple (K, type)	3	3	1

All tested T satisfy $|L_J(F)| = 3^{c(T)}$ byte-exact.

1.5 Significance

This theorem is the structural backbone that makes the Point-Count Theorem possible: because every intersection dimension is frozen at $c(T)$ per level, the inclusion–exclusion of the arrangement is a **fixed finite combinatorial datum**, blind to the tower and to the field. It confines all the arithmetic difficulty of the Sofa problem out of the *lattice of intersections* (which is frozen and torsion-free) and into the single question of how the truncated coordinate ring sits against this frozen lattice — i.e. it is the theorem that lets one say “the torsion does not live in the intersections, only in how the subspaces sum.”

Part of the Sofa Theorem repository: github.com/tretoef-estrella · tretoef@gmail.com